

DB A3 Laiba Fatima 22k-5195

Q1.

1.

$$\pi_{sname}(\sigma_{cname='DatabaseSystems'}(Courses) \bowtie_{Courses.cid=Enrollments.cid} Enrollments \bowtie_{Enrollments.sid=Students.sid} Students)$$

2.

$$\pi_{sid}(\sigma_{grade > 90}(Enrollments))$$

3.

$$\pi_{sname}(Students) - \pi_{sname}(\sigma_{credits < 3}(Courses) \bowtie_{Courses.cid=Enrollments.cid} Enrollments \bowtie_{Enrollments.sid=Students.sid} Students)$$

4.

$$\pi_{sid}(Students) \div \pi_{cid}(Courses)$$

5.

$$\pi_{sname}(\sigma_{count \geq 2}(sname \bowtie count(cid)(Enrollments \bowtie_{Students.sid=Enrollments.sid} Students)))$$

6.

$$\pi_{sname}(Students \bowtie_{Students.sid=Enrollments.sid} (\pi_{cid}(\sigma_{semester='Spring'}(Enrollments)) \div \pi_{cid}(Enrollments)))$$

7.

$$\pi_{sid}(\sigma_{grade > \pi_{grade}(\sigma_{sname='Aliyan'}(Students))}(Students))$$

8.

$$\pi_{sid} \left(\sigma_{grade > \left(\bowtie MAX(grade) \left(\sigma_{sname='Huzaiifa'}(Students) \right) \right)} (Students) \right)$$

9.

$$\pi_{sname}(\sigma_{credits=4}(Courses) \bowtie_{Courses.cid=Enrollments.cid} Enrollments \bowtie_{Enrollments.sid=Students.sid} Students)$$

10.

$\pi_{sid}(Enrollments \bowtie_{Enrollments.cid=Courses.cid} (\sigma_{credits=\$MAX(credits)}(Courses)))$

Q2.

1. $\pi_{title}(\sigma_{inColor = 0 \wedge year > 1970}(Movie))$

Meaning: This retrieves titles of movies that are not in color and were released after 1970.

SELECT title FROM Movie WHERE inColor = 0 AND year > 1970;

2. $\pi_{title}(\sigma_{(studioName = 'MGM' \wedge year > 1970) \vee length < 90}(Movie))$

Meaning: This retrieves titles of movies produced by MGM after 1970 or any movie with a runtime shorter than 90 minutes.

SELECT title FROM Movie WHERE (studioName = 'MGM' AND year > 1970) OR length < 90;

3. $\pi_{title, length}(\sigma_{studioName = 'Disney' \wedge year = 1990}(Movie))$

Meaning: This returns the title and length of movies produced by Disney in 1990.

SELECT title, length FROM Movie WHERE studioName = 'Disney' AND year = 1990;

4. $\pi_{title, length/60}(\sigma_{studioName = 'Disney' \wedge year = 1990}(Movie))$

Meaning: This returns the title and duration (in hours) of Disney movies from 1990.

SELECT title, length/60 AS duration_in_hours FROM Movie WHERE studioName = 'Disney' AND year = 1990;

5. $\pi_{producerC}(Movie \bowtie StarsIn \bowtie \sigma_{starName = 'Harrison Ford'}(StarsIn))$

Meaning: This retrieves producer codes for movies that feature Harrison Ford.

SELECT DISTINCT producerC FROM Movie JOIN StarsIn ON Movie.title = StarsIn.movieTitle AND Movie.year = StarsIn.movieYear WHERE starName = 'Harrison Ford';

6. $\pi_{Star.name, Exec.name}(\sigma_{Star.address = Exec.address}(MovieStar \bowtie MovieExec))$

Meaning: This lists pairs of MovieStar and MovieExec who share the same address.

SELECT MovieStar.name AS StarName, MovieExec.name AS ExecName FROM
MovieStar JOIN MovieExec ON MovieStar.address = MovieExec.address;

7. $\pi_{\text{producerC}}(\sigma_{\text{title} \neq \text{'Star Wars'}}(\text{Movie}))$

Meaning: This retrieves producer codes for all movies except "Star Wars."

SELECT DISTINCT producerC FROM Movie WHERE title != 'Star Wars';

8. $\pi_{\text{producerC}}(\sigma_{\text{studioName} = \text{'Disney'}}(\text{Movie})) \cap \pi_{\text{producerC}}(\sigma_{\text{studioName} = \text{'MGM'}}(\text{Movie}))$

Meaning: This returns producer codes for producers who have worked with both Disney and MGM.

SELECT producerC FROM Movie WHERE studioName = 'Disney' INTERSECT SELECT
producerC FROM Movie WHERE studioName = 'MGM';

9. $\pi_{\text{title}}(\text{Movie}) - \pi_{\text{title}}(\text{Movie} \bowtie \text{MovieExec})$

Meaning: This retrieves titles of movies that do not have an associated MovieExec.

SELECT title FROM Movie WHERE title NOT IN (SELECT title FROM Movie JOIN
MovieExec ON Movie.producerC = MovieExec.cert);

10. $S1 = \rho_{\text{title1}, \text{year1}, \text{name1}}(\text{StarsIn})$

$S2 = \rho_{\text{title2}, \text{year2}, \text{name2}}(\text{StarsIn})$

$(S1) \bowtie \text{name1} = \text{name2} \text{ AND } (\text{title1} \neq \text{title2} \text{ OR } \text{year1} \neq \text{year2}) (S2)$

Meaning: This finds pairs of stars who have acted in different movies.

SELECT S1.starName AS Star1, S2.starName AS Star2, S1.movieTitle AS Movie1,
S2.movieTitle AS Movie2 FROM StarsIn S1 JOIN StarsIn S2 ON S1.starName =
S2.starName AND (S1.movieTitle != S2.movieTitle OR S1.movieYear != S2.movieYear);